

PAPER • OPEN ACCESS

## Laser welding of dissimilar materials - simulation driven optimization of process parameters

To cite this article: M S Serdeczny 2023 *IOP Conf. Ser.: Mater. Sci. Eng.* **1281** 012018

View the [article online](#) for updates and enhancements.

You may also like

- [Experimental and numerical analysis of intermetallics in Al-Mg friction stir welds](#)  
M Naveenkumar, M Subramanian and R Ranjith
- [Temperature field simulation and grain morphology on laser welding-brazing between Ti-6Al-4V and 1050 aluminum alloy](#)  
Xiaohong Zhan, Hengchang Bu, Qiyu Gao et al.
- [A comparative study of friction stir welding and friction stir scribe technique for dissimilar metal joining](#)  
Sapna A Solanki and Anand B Dhruv



**UNITED THROUGH SCIENCE & TECHNOLOGY**

 **The Electrochemical Society**  
Advancing solid state & electrochemical science & technology

**248th  
ECS Meeting**  
Chicago, IL  
October 12-16, 2025  
*Hilton Chicago*

**Science +  
Technology +  
YOU!**

**SUBMIT  
ABSTRACTS by  
March 28, 2025**

**SUBMIT NOW**

# Laser welding of dissimilar materials – simulation driven optimization of process parameters

**M S Serdeczny**

Flow Science Inc., 683 Harkle Road, Santa Fe, NM 87505, US

marcin.serdeczny@flow3d.com

**Abstract.** Laser welding can be used to join dissimilar materials to produce lightweight structures, and electric vehicle battery systems, which are important means of limiting the carbon emissions in the transport industry. Due to the differences in melting temperatures, thermal conductivities, and mutual solubility of dissimilar materials, it is still challenging to create defect-free joints with high mechanical strength, or low contact electrical resistance. In this work, we present a state-of-the-art numerical model of laser welding, developed within the Computational Fluids Dynamics (CFD) paradigm. The multi-physics model simulates melting, flow, and solidification of the alloys and accounts for the laser-material interactions, phase change, temperature and alloy-dependent thermophysical properties, recoil pressure, buoyancy force, and Marangoni effect. The simulation predicts weld penetration depth and width, alloy mixing, as well as the temperature gradient and cooling rate during the solidification that can be further fed into a micro-structure prediction model. The model is coupled with an optimization tool, which iterates over different process parameters to optimize the joint. The methodology is presented for steel to aluminium welding in lap configuration, but it can be used for other materials such as steel-copper, aluminium-copper, or steel-nickel, and in arbitrary geometrical configuration.

## 1. Introduction

Joining dissimilar materials efficiently and robustly is important for the automotive industry, as numerous parts have complex functions that require locally tailored material properties. Laser welding offers high efficiency, high precision, large throughput, and narrow heat affected zone. To this day, one of the main challenges of welding dissimilar materials are differences in the thermophysical properties. Among the widely used alloys, steel and aluminium couple is one of the most difficult to join via laser welding [1]. Steel has significantly higher melting point, lower thermal conductivity, and lower thermal expansion coefficient than aluminium. On the hand, aluminium has lower laser absorption rate that makes it difficult to select optimal parameters in the process. Moreover, an additional challenge is the formation of intermetallic phases (IMCs) that significantly reduce the electrical and mechanical performance of the joint [2]. IMCs formation depends on the dynamics of the dissimilar materials mixing as well as the temperature history of the weld. Thus, the optimization of laser welding parameters to ensure a proper bond while minimizing the formation of IMCs is an important research area.

Key tools in the analysis of laser welding of dissimilar materials are scanning electron microscopy (SEM), tensile tests, and measurements of the microhardness profile across the seam, which are all



performed after the welding process is finished. In [3], Torkamany et al. looked at the dissimilar welding of carbon steel to aluminium with Nd:YAG pulsed laser. The influence of the laser peak power, pulse duration, and welding speed on the material mixing and IMCs formation was studied. It was found that too high line energy (high peak power, long pulse, or low welding speed) will cause significant mixing of the aluminium with steel and inevitable formation of the brittle IMCs as well as the crack propagation. This was found to be correlated with the low tensile strength of joints produced with high line energy input. On the other hand, when the line energy input was low, the shallow penetration of the laser resulted in low tensile strength, even though the IMCs content was minimal. Mathieu et al. [4] used a continuous wave Nd:YAG laser with 3.5 kW power to weld a 6016 T4 aluminium to a low carbon steel. They used an overlap joint configuration and applied a filler wire made of 85% Zn and 15% Al. They presented weld fracture locations and showed that the macroscopic geometry of the weld is equally important as the presence of brittle IMCs. Pereira et al. [1] demonstrated successful welding of DP1000 steel to Al050 in lap configuration but reported that in the butt configuration, the two materials were nearly impossible to weld. Sun et al. [5] added an aluminium ER4043 filler wire in the butt configuration and demonstrated successful welds of AA6013 with low carbon mild steel using laser welding-brazing mode. They investigated the influence of the laser power, wire speed, and the deviation of the laser spot with respect to the aluminium edge on the seam geometry, microstructure, IMCs layer thickness, as well as the tensile strength of the joint. With the increasing line energy density, they observed better wetting and penetration of the weld. Nevertheless, the specimens with higher line energy density yielded a smaller tensile strength due to a thicker layer of IMCs. The formation of intermetallic phases was attributed to the diffusion process of Fe into the melted filler material, as the diffusion coefficient of Fe increases with temperature. To sum up the experimental efforts, it can be noted that to obtain a high strength joint the heat input needs to be optimized to ensure a thin layer of IMCs while the overall seam dimensions need to be maximized. For more experimental work on dissimilar materials welding, the reader is referred to the recent review by Sadeghian and Iqbal [2].

In contrast to most of the experimental methods, the numerical models allow for an in-situ analysis of the process, where the influence of physical phenomena on the material mixing can be studied individually and in detail, allowing for explanatory approach. Moreover, a validated numerical tool would allow for process parameters optimization without costly experimental trials. Several studies have focused on the modelling of the laser welding of dissimilar materials in the conduction mode [6, 7, 8, 9], where the surface of the material is typically assumed flat during the simulation. Keyhole simulations are more difficult to perform, as the free surface tracking algorithm is required, as well as the calculations of the evaporation pressure that drive the melt pool flow. Keyhole dynamics was studied using an-house Computational Fluid Dynamics (CFD) code by Kouraytem et al. [10] for a stainless steel 304. Esfahani et al. [6] simulated mixing of stainless steel with low carbon steel and found it to be influenced by the Marangoni flows that originate from the temperature dependent surface tension. In their model, the recoil pressure was neglected. Recently, Huang et al. [11] investigated numerically the mixing of aluminium and copper during the laser keyhole welding in a lap configuration. They showed how increasing the laser power relates to a larger recoil pressure and higher velocity inside the melt pool and therefore more efficient mixing of dissimilar materials. On the other hand, increasing the weld speed, reduces the time spent on accelerating the fluid by the recoil force, which limits the mixing capacity.

In this work, we study the pulsed laser welding of steel to aluminium using a CFD model. The pulsed laser is used to precisely control the heat input through the pulse duration, frequency, and peak power. First, a state-of-the-art model is developed and validated with experimental measurements. In the second part, the CFD model is coupled with an optimization algorithm that iterates over process parameters and minimizes the material mixing, while ensuring the maximum penetration of the material and good bonding. Section 2 describes the CFD model, simulation parameters, and the framework in which the model is coupled with the optimization tool. In Section 3, the results of the CFD simulations are shown. First, the agreement of the numerical results with the experimental measurements is discussed. Next, the influence of process parameters on the material mixing is investigated. At the end of Section 3, the optimization results are presented. Section 4 summarizes and concludes the study.

## 2. Methodology

### 2.1. Physics of the numerical model

In this work, the laser welding is simulated using a commercial CFD solver **FLOW-3D WELD** release 7 update 2 [12]. The core of the computational model is the solver for the mass, momentum and energy conservation equations that can be written in a simplified form as

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot (\mu \bar{\bar{\mathbf{S}}}) + \rho \mathbf{g} \quad (2)$$

$$\rho \frac{DI}{Dt} = \nabla \cdot (k \nabla T) + Q_L \quad (3)$$

$$\bar{\bar{\mathbf{S}}} = \nabla \mathbf{u} + \nabla \mathbf{u}^T \quad (4)$$

where  $\mathbf{u}$  is the velocity vector,  $\rho$  is the material density,  $D()/Dt$  is the notation of the material derivative,  $t$  is the time,  $p$  is the pressure,  $\mu$  is the dynamic viscosity,  $\bar{\bar{\mathbf{S}}}$  is the strain rate tensor,  $\mathbf{g}$  is the gravitational acceleration,  $I = C_v(T)$  is the internal energy of the material,  $C_v$  is the specific heat,  $T$  is the temperature,  $k$  is the thermal conductivity, and  $Q_L$  is the laser heat source.

The above system of equations is supplemented by the Volume-Of-Fluid (VOF) method that tracks the position of the material free surface with respect to the atmosphere [13]. The VOF method relies on solving an equation for variable  $F$ , called fraction of fluid.

$$\frac{\partial F}{\partial t} + \mathbf{u} \cdot \nabla F = 0 \quad (5)$$

Fraction of fluid takes values of 1 for the phase 1 (both liquid and solid matter), and value 0 for the phase 2, i.e. the atmosphere. In **FLOW-3D WELD**, the conservation equations (1) - (4) are solved only for phase 1. Phase 2 is approximated with a constant pressure and temperature and no velocity. This approach greatly enhances the computational time. The dissimilar alloys are tracked as a sub-phases of the phase 1, and their concentration is calculated by solving a scalar transport equation

$$\frac{\partial C}{\partial t} + \mathbf{u} \cdot \nabla C = \nabla \cdot (\phi \nabla C) \quad (6)$$

where  $C$  is the marker variable for alloys: 0 for steel and 1 for aluminium;  $\phi$  is the diffusion coefficient that is calculated as

$$\phi = Sc \cdot \mu / \rho \quad (7)$$

where  $Sc$  is the Schmidt number that was assumed equal 1. Wherever the concentration  $C$  is between 0 and 1, the law of mixture is used to approximate the material properties (the viscosity, thermal conductivity, specific heat etc.).

Apart from the above, the following physical phenomena are included in the model:

- Density dependency on the temperature is included in the momentum conservation equation to capture buoyancy driven flows inside the melt pool.
- Surface Tension is modelled through an additional pressure term on the free surface

$$P_{ST} = \sigma(T) \cdot \kappa \quad (8)$$

where  $\sigma$  is the surface tension coefficient that depends on the temperature. The dependency of  $\sigma$  on  $T$  allows to capture the Marangoni flows, where typically low temperature regions have higher surface tension and pull the fluid from the hot regions with low surface tension.  $\kappa$  is the curvature of the free surface at a given point on the material surface.

- Solidification phenomenon is included in the energy and momentum conservation by accounting for the latent heat of the phase change as well as the additional drag on the flow in the mushy zone (during the crystallization and melting). The velocity field is considered frozen in the solidified fluid cells.
- Laser absorption was approximated as constant, while in reality it is temperature and angle dependent. However, obtaining a reliable data is difficult. The starting values for low carbon steel and 5754 aluminum were  $\varphi_{St14} = 0.35$  and  $\varphi_{Al5754} = 0.25$ , respectively. The values were further fine-tuned, when comparing the simulation results with experimental measurements, which is described in Section 3.1.
- A ray tracing method was used to track the reflections of the laser beam and subsequent absorption. The method allows to capture the effective increase of absorptivity in a keyhole scenario, i.e. when laser rays are trapped in the material due to multiple reflections. The ray tracing method works by dividing the laser beam into rays that are stored in the computational cells. When a ray interacts with the material, the incident and reflection angles are computed, as well as the portions of the absorbed and reflected energy. In the model, up to 7 reflections of a single ray are tracked, which is typically when the energy carried by a single ray becomes very small.
- The shield gas dynamic pressure was included in the simulation as

$$p_{sg} = \frac{1}{2} \rho_{sg} u_{sg}^2 \quad (9)$$

where  $\rho_{sg}, u_{sg}$  are the density and velocity of the shielding gas. Argon density and  $u_{sg} = 1.0 \frac{m}{s}$  were used to follow the experimental measurements used to calibrate the model.

- Liquid-Vapor phase change is modelled by including its effect in mass, energy, and momentum conservation of phase 1. Mass loss due to material evaporation is calculated according to the following equation [14]

$$\dot{m} = \alpha \sqrt{\frac{1}{2\pi R_{gas} T_{liq}^{inter}}} (P_{sat}(T) - P_{vap}) \quad (10)$$

where  $R_{gas}$  is the specific gas constant,  $T_{liq}^{inter}$  is the average temperature at the liquid-vapor interface,  $P_{sat}(T)$  is the saturation pressure at the local temperature,  $P_{vap}$  is the partial pressure of the metal phase in the atmosphere (constant),  $\alpha$  is the accommodation coefficient and it expresses the probability that the molecule escapes the liquid phase to the vapor phase. In practice, for alloys, this coefficient is unknown and must be chosen based on the experimental observations. In this work, we use a value of 0.01 based on the previous experience in numerical simulations of evaporation of steel and aluminum alloys.

Evaporation mass flux results in a change of the internal energy of the metal (cooling effect) due to the latent heat of evaporation

$$\Delta I = \Delta m \cdot \lambda_{evap} \quad (11)$$

$\Delta I$  is the change of internal energy,  $\Delta m$  is the change of mass due to evaporation of the material,  $\lambda_{evap}$  is the latent heat of evaporation.

The third implication of the liquid-vapor phase change is the evaporation (or recoil) pressure, which is due to the momentum of the molecules leaving the liquid state and rapidly expanding into the gaseous state. The evaporation pressure in **FLOW-3D WELD** is modelled with the following equation

$$P_r = AP_v \exp \left( B \left( 1 - \frac{T_v}{T_s} \right) \right) \quad (12)$$

where  $P_v, T_v$  are coordinates of a single point on the saturation curve,  $T_s$  is the average temperature at the liquid-vapor interface, and  $A, B$  are the model parameters. In several works [15, 16, 17],  $A$  is assumed to be equal 0.54, following the early works of Anisimov, which is used as a starting point in this work. The coefficient  $B$  can be estimated by comparing Eq. (12) to the Clausius-Clapeyron equation

$$B = \frac{\lambda_{evap}}{(\gamma - 1)c_v T_v} \quad (13)$$

where  $\gamma$  is the specific heat ratio.

## 2.2. Geometry of the model

Figure 1 depicts the geometry of the welding configuration, as implemented in the numerical model. The geometry follows the experimental measurements of Torkamany et al. [3] and Ghaini et al. [18], which will be reproduced as a validation case. The lap configuration consists of a low carbon steel St14 plate (blue block in Figure 1) on top of an aluminium 5754 plate (red block in Figure 1). The thickness of the steel plate is 0.8 mm, except the validation cases B-1 to B-5 (Section 3.1. where it is equal 0.7 mm). The thickness of aluminium plate is 2 mm, except cases B-1 to B-5 where only the steel plate is used. The width and length of the plates in the model are 20 mm and 25 mm, respectively, which is considerably smaller than in the actual measurements but necessary to make the simulation time feasible. The plates are stationary while the laser beam moves at a constant velocity  $V$  in x-direction.

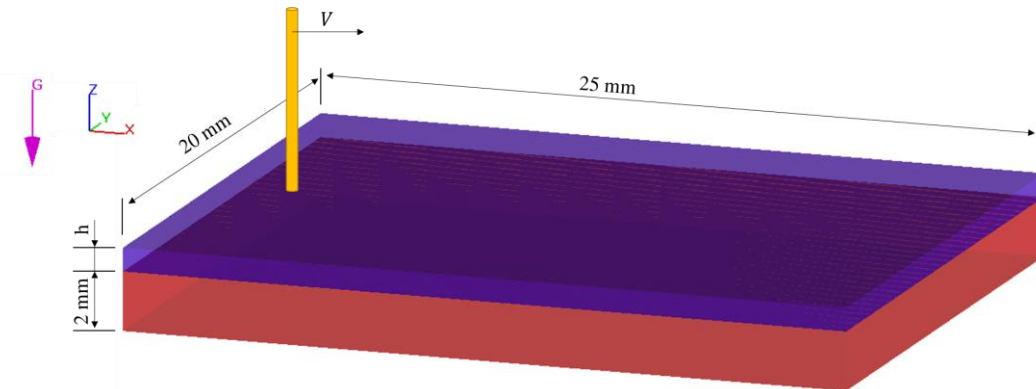


Figure 1. Geometry of the welding configuration implemented in the numerical model. Blue region is the low carbon steel St14, while the red region is the aluminium alloy 5754.

## 2.3. Mesh

Figure 2 shows the cartesian mesh that is used in the simulation. The mesh extends above and below the workpieces in z-direction by 0.5 mm and aligns with the plates dimensions in x and y. The mesh is made of two blocks: 1) Inner mesh block that serves to resolve the fluid flow inside the melt pool and heat transfer to the solidified regions in the vicinity of the melt pool; 2) Outer mesh block that serves to resolve the heat transfer in the vicinity of the domain boundaries. The details about the cell size and cell count are given in Table 1. Three levels of mesh refinement were tested (coarse, medium, and fine) and their influence on the melt pool dimensions and material mixing was checked. The results of the mesh sensitivity analysis are shown in Appendix A in Figure A.1., where the melt pool shape and dimensions (depth and width) in the YZ-cross-section can be seen. The melt pool dimensions were found to vary mildly between the medium and the fine mesh, while the computation time increases from approx. 5-10 hours to 2-3 days (depending on the case). There is, however, a difference in the alloy composition shown with the colouring variable, and for the fine mesh, more significant upward motion of the



aluminium is observed. To limit the computational effort, we use the medium mesh for initial screening and for optimization, while the final validation cases are re-simulated with the fine mesh to confirm the findings.

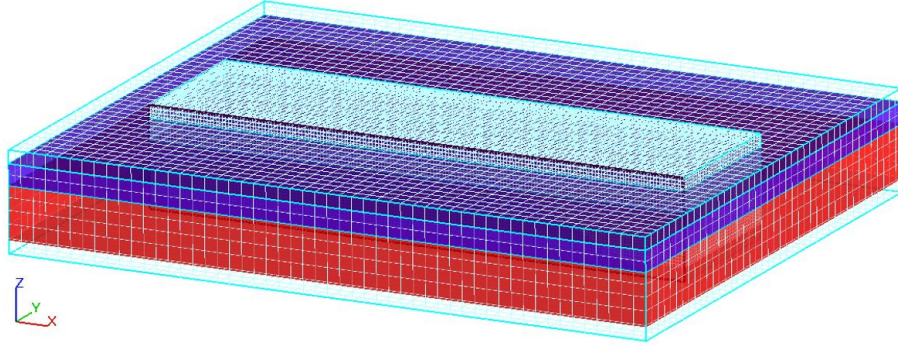


Figure 2. Cartesian mesh used in the simulations. The mesh consists of two blocks: inner – with small cell size, and outer that is coarser.

Table 1. Mesh properties.

|            | Outer Mesh Block  |                   |                   | Inner Mesh Block  |                   |                  |
|------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|
|            | Coarse            | Medium            | Fine              | Coarse            | Medium            | Fine             |
| Cell size  | 960 $\mu\text{m}$ | 480 $\mu\text{m}$ | 240 $\mu\text{m}$ | 240 $\mu\text{m}$ | 120 $\mu\text{m}$ | 60 $\mu\text{m}$ |
| Cell count | 2.7e3             | 32e3              | 138e3             | 34.8e3            | 280e3             | 2.2e6            |

#### 2.4. Initial and boundary conditions

The four vertical boundaries of the computational domain were prescribed a fixed temperature of 500 K. The initial temperature of the material as well as the ambient atmosphere temperature were also set to 500 K. A sensitivity analysis of the results with respect to this temperature was performed (at 400 K and 600 K) and no significant change in the melt pool dimensions, or mixing was observed. The top and bottom boundaries of the computational domain were prescribed a fixed pressure of one bar and the temperature of 500 K.

As the motion of the gas phase above the metal surface was not calculated explicitly, the convective heat loss was modelled by setting the following condition at the metal free surface

$$Q = h(T_s - T_\infty) \quad (14)$$

where  $h$  is the heat transfer coefficient (assumed 30 W/m<sup>2</sup>K which is in the range of forced convection, that should be driven by the shielding gas [19]).  $T_s$  is the temperature at the free surface,  $T_\infty$  is the ambient gas temperature (assumed 500 K). The energy convected from the metal to the atmosphere is lost.

#### 2.5. Laser parameters

The laser beam source was assumed to have a Gaussian intensity profile and have a cylindrical shape (no divergence or convergence of the beam)

$$H(r) = H_c \exp\left(-2\left(\frac{r}{W}\right)^2\right) \quad (15)$$

where  $H_c$  is the heat flux in the centre of the beam,  $r$  is the radial coordinate,  $W$  is the laser spot radius, defined as a point when the intensity (heat flux) falls below  $1/e^2 \approx 0.135$  of the maximum intensity, which corresponds to two standard deviations of the normal distribution. For a Gaussian beam, this definition is equivalent to  $D4\sigma$ , where the beam diameter is measured by the 4 standard deviations.

## 2.6. Material parameters

The chemical compositions of the two alloys are presented in Table 2. The thermophysical properties were generated using JMatPro software [20] and included the temperature dependency of viscosity, density, specific heat, thermal conductivity and surface tension.

Table 2. Chemical compositions of the alloys used in this work, in wt% [3].

| Element | Si   | Cu   | Mn   | Mg    | Cr   | Ni    | C   | Fe   | Al   |
|---------|------|------|------|-------|------|-------|-----|------|------|
| Al 5754 | 0.12 | 0.04 | 0.25 | 3.009 | 0.08 | 0.006 | -   | 0.37 | Bal. |
| St 14   | -    | -    | 0.4  | -     | -    | -     | 0.4 | Bal. | -    |

## 2.7. Optimization tool

CFD model was coupled with an optimization tool **FLOW-3D(x)** 2022R1 that is a general-purpose optimization algorithm that employs hybrid and surrogate approach. The hybrid approach means that the algorithm will switch between the local or global minimum search depending on the number of iterations that are left till the end of the optimization (prescribed by the user). Surrogate model is particularly suitable for computationally expensive problems, such as CFD or Finite Element Method (FEM) and relies on reproducing the real objective function with an acceptable surrogate function that is cheap to evaluate. The surrogate function is used during the optimization process to select points at which the real objective function will be evaluated in the next iteration.

Figure 3 shows a simplified schema of a single iteration of the optimization tool. In Step 1, the guessed peak power and pulse duration of the laser are passed to a Python script that generates a .csv file, which contains a time series of square shape pulses. Next node, in Step 2, is the **FLOW-3D WELD** node, i.e. the CFD solver. The path of the csv file as well as the guessed value of welding speed are inputted to the CFD solver node, which runs the welding simulation. In Step 3, the solver node feeds the path of the simulation results to the execution node that runs a second Python script, which launches a CFD post-processor, **FLOW-3D POST**, and extracts the average melt pool penetration depth, as well as the average aluminium content. The average aluminium content is extracted in the weld region above the steel-aluminium interface, so it quantifies the amount of aluminium that flowed upwards due to the keyhole-driven mixing process. In Step 4, based on the surrogate objective function, algorithm guesses new values of peak power, peak duration, and welding speed, in order to maximize the penetration and minimize the aluminium content. The cycle repeats with the new input.

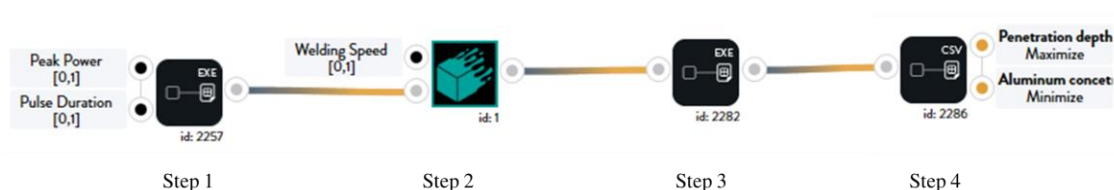


Figure 3. A diagram representing a single iteration of the optimization model in **FLOW-3D (x)**.

## 3. Results

### 3.1. Validation of the numerical model

We start by calibrating the model to the measurements of Torkmany and Ghaini et al. presented in the series of publications [21] [18] [3]. The first set of measurement data is summarized in Table 3 (B-1 to B-5, following the measurement numbering of [21] [18]) and consisted of a single pulsed laser scan on a 0.7 mm thick plate of low carbon steel St14. This is a simplified geometry with respect to Figure 1, as there is no second material (aluminium plate) present yet in the simulation domain.



Table 3. Process conditions used in the validation of the numerical model. The conditions are taken from: B-1 to B-4 [20] [21], TN14 - TN19 [3]. In all cases, the pulse frequency was 20 Hz.

| Simulation /Experiment | Peak power, $P_p$ [kW] | Pulse duration, $P_d$ [ms] | Welding speed, $V$ [mm/s] | Laser spot diameter, $2W$ [mm] |
|------------------------|------------------------|----------------------------|---------------------------|--------------------------------|
| B-1                    | 1.33                   | 6                          | 8.3                       | 1.2                            |
| B-2                    | 1.50                   | 6                          | 8.3                       | 1.2                            |
| B-3                    | 1.67                   | 6                          | 8.3                       | 1.2                            |
| B-4                    | 1.83                   | 6                          | 8.3                       | 1.2                            |
| B-5                    | 2.00                   | 6                          | 8.3                       | 1.2                            |
| TN-14                  | 1.43                   | 5                          | 11.5                      | 0.76                           |
| TN-15                  | 1.43                   | 5                          | 10                        | 0.76                           |
| TN-16                  | 1.43                   | 5                          | 8                         | 0.76                           |
| TN-17                  | 1.43                   | 5                          | 6                         | 0.76                           |
| TN-18                  | 1.43                   | 5                          | 4                         | 0.76                           |
| TN-19                  | 1.43                   | 5                          | 2                         | 0.76                           |

The comparison of the simulated and measured width and depth of the melt pool is shown in Figure 4. Two simulation runs were performed – one with the nominal absorptivity of 0.35, and with increased absorptivity of the material to 0.45. The rationale of varying material absorptivity to laser radiation is due to unknown surface conditions of the material in the experiment (surface roughness, and oxides layer), which may significantly influence the reflectivity and absorptivity [22] [23]. Moreover, the absorptivity is approximated as constant in the model, while it varies with the angle of the incidence and temperature.

The measured melt pool width first increases with the laser peak power, and then remains constant. The simulations with nominal absorptivity  $\varphi = 0.35$  capture this trend more accurately than when the absorptivity is increased. On the other hand, the melt pool depth is underestimated when  $\varphi = 0.35$ , hence the model was run with  $\varphi = 0.45$ . It is noted that in these simulations, the recoil pressure model coefficient  $A$  was increased to 2, as the default value of 0.54 (see Section 2.1. ) yielded significantly smaller material penetration and no transitioning to the keyhole. Increase of coefficient  $A$  above 2 had small impact on the penetration depth, as the heat absorbed by the material is a major driver for the evaporation process and the recoil force. As the accurate prediction of the penetration depth is prioritized, absorptivity value of 0.45 is used in the following simulations.

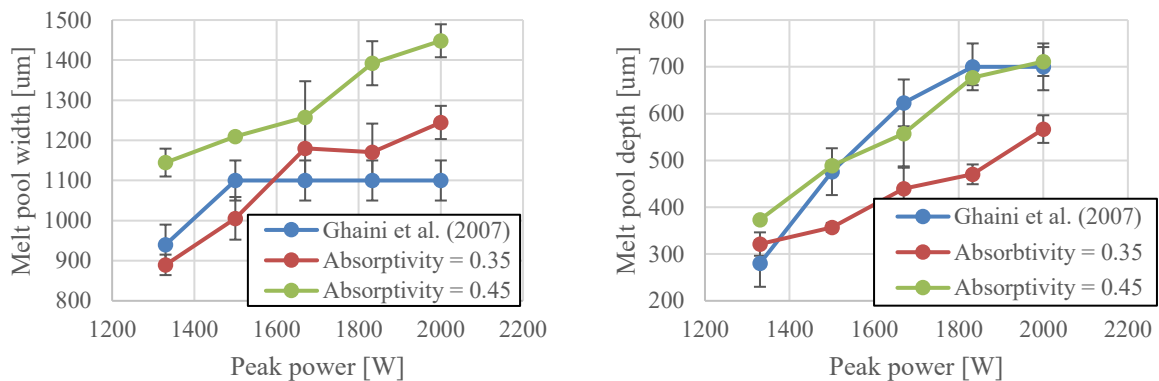


Figure 4. Validation of the numerical model to the experimental measurements of Ghaini et al. [21] Left: Melt pool width. Right: Melt pool depth. For each simulation point, the width and depth were calculated at three cross-sections along the weld ( $x = 5$  mm, 7.5 mm and 10 mm) and averaged. The standard deviation from the 3 calculations is shown with the error bar. For the measurements of Ghaini et al. [21], 50  $\mu\text{m}$  error bar was assumed.

Figure 5 shows the cross-sections of the melt pools obtained with the numerical model (light blue colour) for the absorptivity value of 0.45 and photographed by Torkamany et al. [21] (black dashed line). The overall dimensions of the simulation and measured melt pools are similar but there are differences in the shape. The simulated cross-sections feature a concave surface that indicates strong action of the recoil pressure, while the measured cross-sections are rather flat. Moreover, the measured melt pool shapes are more triangular and pointier. This could be due to the differences in the laser spot diameter between the experiment and the simulation that could stem out of either measurement error, or inconsistent definitions of the beam diameters. Here, we used four standard deviations of the Gaussian distribution to define the beam diameter, while the exact definition is not mentioned in [21]. Alternatively, the intensity profile of the laser beam could be different than the Gaussian that was assumed in the numerical model.

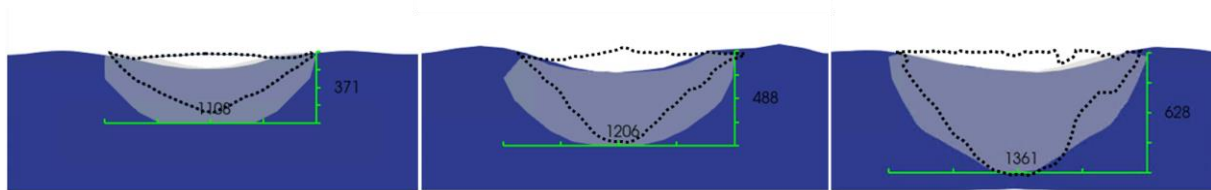


Figure 5. Comparison of the simulated (light blue) and measured [21] (black dashed line) melt pool cross-sections.

In the next step, the modelling results are compared to the melt pool measurements during welding of steel and aluminium [3] that are considered more complex to reproduce. Table 3 summarizes the welding parameters used in the experiment and to run the simulations in this round (TN-14 to TN-19). Figure 6 presents the comparison between the measured and simulated melt pool dimensions as a function of welding speed. First, the model is run with absorptivity value 0.45, that was used in the previous simulation round. It can be seen that the width of the melt pool is captured accurately at high welding speeds (low line energy density), but it is underestimated at lower welding speeds. At the same time, the melt pool depth is underestimated in the entire range of welding speeds indicating that either too little energy is absorbed by the material, or the recoil pressure is underestimated and does not lead to sufficient penetration of the material (in this simulation round the recoil pressure coefficient was increased to  $A = 10$ ).

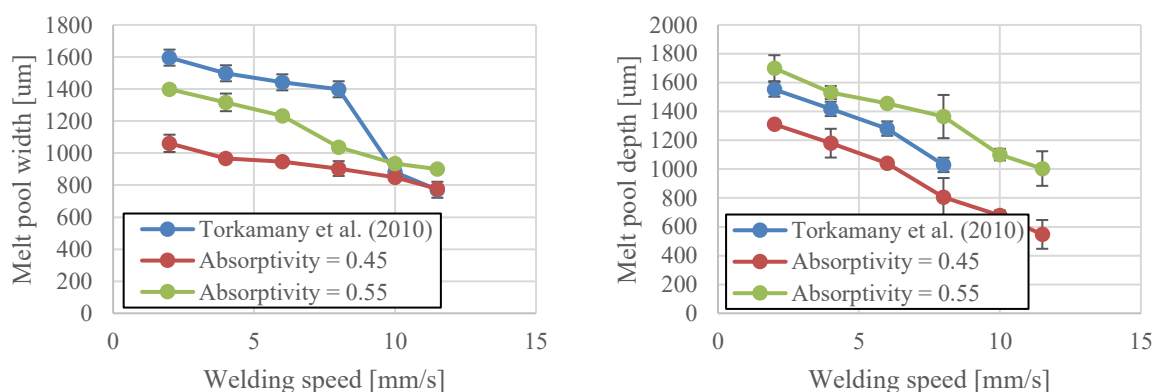


Figure 6. Validation of the numerical model to the experimental measurements of Torkamany et al. [3]. Left: Melt pool width. Right: Melt pool depth. For each simulation point, the width and depth were calculated at three cross-sections along the weld ( $x = 5$  mm,  $7.5$  mm, and  $10$  mm) and averaged. The standard deviation of the 3 calculations is shown with the error bar. For the measurements of Torkamany et al. [3],  $50$   $\mu\text{m}$  error bar was assumed.

The melt pool cross-sections simulated with absorptivity value of 0.45 and  $A = 10$  are compared with the measurements in Figure 7. The measured weld becomes deeper with decreasing welding speed, as the heat input per length increases. The model captures this dependency although the penetration depth is smaller than in the experiment. Based on the measured profile, it can be seen that the weld shape becomes more irregular at the higher powers, which is due to unstable keyhole behaviour. The simulated weld shape features a widening, close to the interface between the two alloys. This widening is mildly seen in the experiment when  $V = 4$  mm/s.

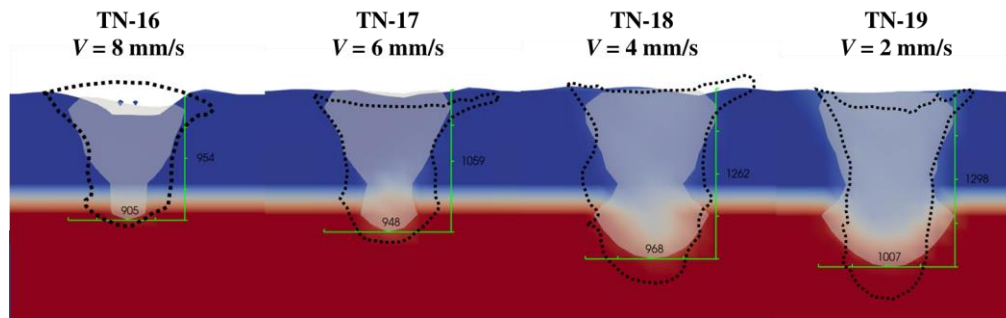


Figure 7. Comparison of the simulated (light blue region) and measured (black dashed line) melt pool cross-sections. Dark blue indicates steel, while red indicates aluminium. For simulations TN-14 and TN-15, the penetration was too shallow to create a joint (both in the simulation and experiment).

### 3.2. Analysis of the material mixing

In this section, we look on how the process conditions influence the mixing of the alloys. Figure 8 shows a cut through the weld in  $XZ$ -plane. On the left, simulation performed with welding speed of 2 mm/s is shown, while on the right with 6 mm/s. At low welding velocity, larger concentration of aluminium is observed in the weld, which agrees with the experimental observations in [3]. This is contributed to the higher energy density, and therefore larger recoil pressure and longer time that the laser acts on the material, which allows for larger accelerations of the molten aluminium. Clusters of aluminium are observed to be lifted upwards in the weld due to a vortex that forms behind the keyhole [11]. In Figure 9, aluminium volume fraction was plotted along the weld. The observed periodic spikes in the aluminium content can be regions of elevated probability of IMCs formations.

Figure 10 shows the aluminium volume fraction in the weld region above the steel-aluminium interface. In other words, it measures the aluminium volume that flowed upwards due to the keyhole mixing. Aluminium volume fraction falls almost linearly with the increasing welding speed until the heat input per length is insufficient to create a keyhole and the melt pool does not reach the Al plate.

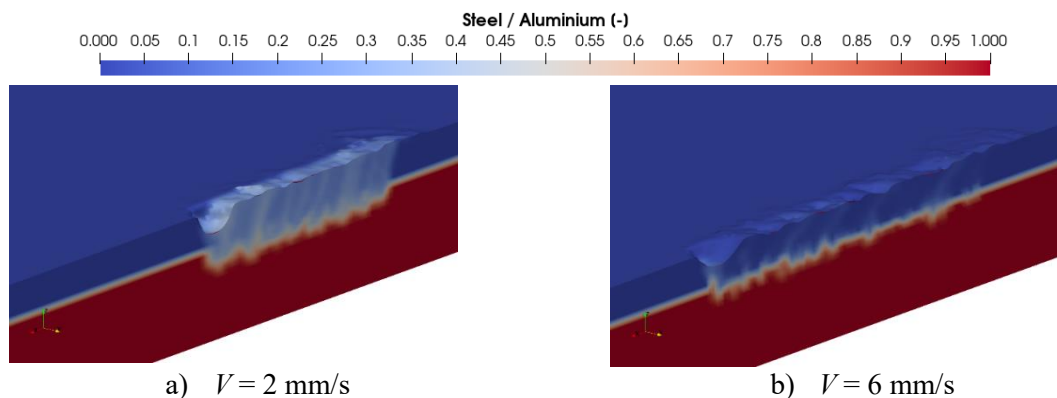


Figure 8. Cross-section through  $XZ$  plane that shows alloys mixing simulated for two welding speeds. Other parameters were set as in Section 3.1.

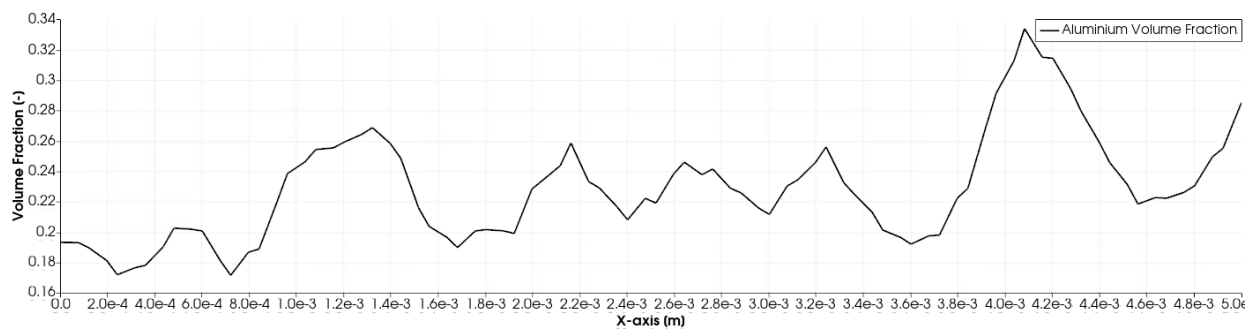


Figure 9. Variation in aluminium concentration along the weld at  $Z = -500 \mu\text{m}$  and  $Y = 0 \mu\text{m}$  for  $V = 2 \text{ mm/s}$ .

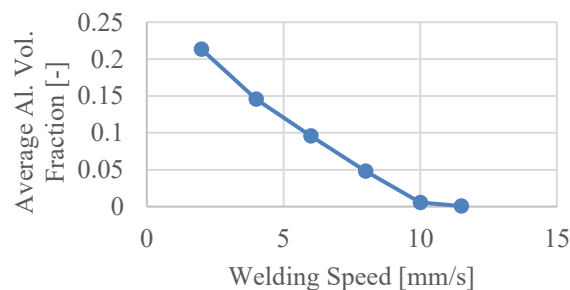


Figure 10. Dependency of the average aluminium volume fraction in the weld region above steel/aluminium plates' interface.

### 3.3. Laser welding optimization results

In this part, the calibrated model was run using the optimization tool described in Section 2.7. The algorithm was set to maximize the penetration depth and minimize the aluminium content in the upper part of the weld, while varying the peak power, pulse duration, and welding velocity. The model was allowed to run 7 iterations on a single 10 core machine, which took around 14 hrs. The preliminary results are shown in Figure 11, where each point is a single iteration. Y-axis shows the mean penetration depth, X-axis shows the laser peak power, while the bubble size indicates the average aluminium content in the weld region. Generally, the average aluminium content increases with the penetration depth, which makes it a very difficult optimization task. However, some of the trends in Figure 11 indicate that by a proper choice of process conditions, improvements are possible. For instance, as indicated by the black arrow, the mean penetration depth could be increased from approx. 1.2 mm to 1.5 mm, while keeping almost the same aluminium volume fraction in the weld, which yields larger joint area without increasing the risk of brittle IMCs formation.

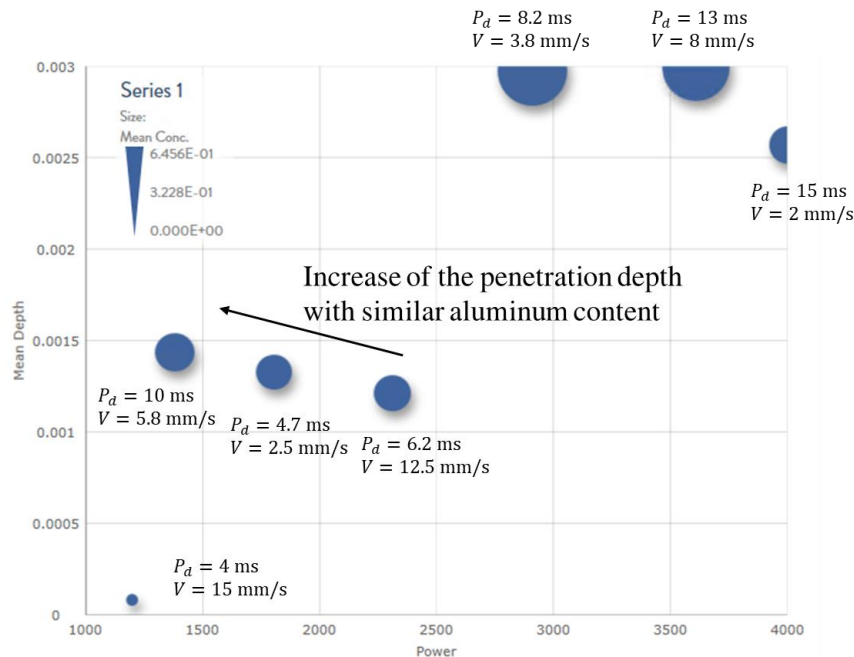


Figure 11. Preliminary optimization results, obtained for 7 iterations.

#### 4. Summary and conclusions

This work documents a development of a computational model for simulating the pulsed laser welding of steel to aluminium. The model relies on solving the conservation equations of mass, momentum, and energy. Moreover, the method uses an empirical equation to predict the recoil pressure to capture the keyhole regime. The model was calibrated to the experimental measurements by comparing the simulated and measured melt pool cross-sections. As calibration parameters, material absorptivity and recoil pressure coefficients were used. However, for the chosen set of process conditions, the numerical results were primarily found to be sensitive to the absorptivity. The model correctly predicts the trends in the melt pool dimensions when the heat input per length is increased (increase in peak power or decrease in welding speed). After the calibration, the simulated melt pool shapes presented a good agreement with the measured melt-pool cross-sectional shapes, considering the complexity of the keyhole dynamics. The discrepancies were attributed to the uncertainty of the beam diameter definition in the experiment, unknown laser beam intensity profile, as well as simplistic recoil pressure model.

In the second part of the work, aluminium and steel mixing was observed as a function of welding speed. It was found that the clusters of aluminium flow upward due to the keyholing and lead to ununiform aluminium content in the weld with periodic spikes that can potentially be sources of brittle intermetallic phases formations.

In the third part, the calibrated CFD model was coupled with an optimization algorithm that iterates over the peak power, peak duration, and welding velocity to maximize the penetration depth, and minimize the aluminium content in the seam that increases the risk of IMCs formation. So far, the preliminary results indicated that the incremental improvements are possible, although the positive correlation between the aluminium content and the penetration depth, makes this optimization task difficult.

## A. Appendix

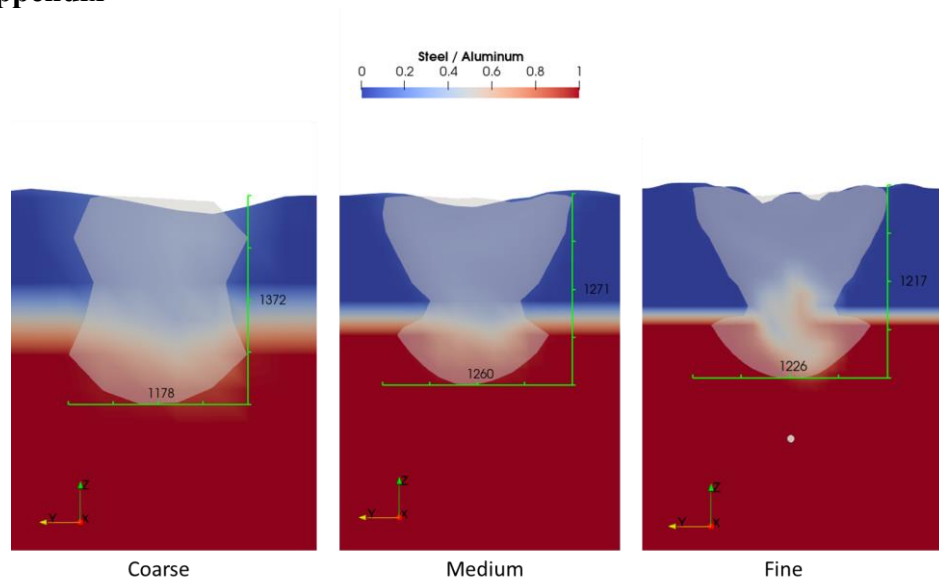


Figure A.1. Mesh convergence study results.

## References

- [1] A. B. Pereira, A. Cabrinha, F. Rocha, P. Marques, F. A. O. Fernandes and R. J. A. d. Sousa, "Dissimilar Metals Laser Welding between DP1000 Steel and Aluminum Alloy 1050," *Metals*, vol. 9, no. 102, 2019.
- [2] A. Sadeghian and N. Iqbal, "A review on dissimilar laser welding of steel-copper, steel-aluminum, aluminum-copper, and steel-nickel for electric vehicle, battery manufacturing," *Optics and Laser Technology*, vol. 146, p. 107595, 2022.
- [3] M. Torkamany, S. Tahamtan and J. Sabbaghzadeh, "Dissimilar welding of carbon steel to 5754 aluminum alloy by Nd:YAG pulsed laser," *Materials and Design*, vol. 31, p. 458–465, 2010.
- [4] A. Mathieu, R. Shabadi, A. Deschamps, M. Suery, S. Mattei, D. Grevey and E. Cicala, "Dissimilar material joining using laser (aluminum to steel using zinc-based filler wire)," *Optics & Laser Technology*, vol. 39, pp. 652-661, 2007.
- [5] J. Sun, Q. Yan, W. Gao and J. Huang, "Investigation of laser welding on butt joints of Al/steel dissimilar materials," *Materials & Design*, vol. 83, pp. 120-128, 2015.
- [6] M. N. Esfahani, J. Coupland and S. Marimuthu, "Numerical simulation of alloy composition in dissimilar laser welding," *Journal of Materials Processing Technology*, vol. 224, pp. 135-142, 2015.
- [7] Z. Xue, S. Hu, D. Zuo, W. Cai, D. Lee and E. K.-A. Jr, "Molten pool characterization of laser lap welded copper and aluminum," *J. Phys. D: Appl. Phys.*, vol. 46, no. 495501, p. 9, 2013.
- [8] Y. Hu, X. He, G. Yu, Z. Ge, C. Zheng and W. Ning, "Heat and mass transfer in laser dissimilar welding of stainless steel and nickel," *Applied Surface Science*, Vols. 5914-5922, p. 258, 2012.
- [9] S. Zhao, G. Yu, X. He, Y. Zhang and W. Ning, "Numerical simulation and experimental investigation of laser overlap welding of Ti6Al4V and 42CrMo," *Journal of Materials Processing Technology*, vol. 211, pp. 530 - 537, 2011.



- [10] N. Kouraytem, X. Li, R. Cunningham, C. Zhao, N. Parab, T. Sun, A. D. Rollett, A. D. Spear and a. W. Tan, "Effect of Laser-Matter Interaction on Molten Pool Flow and Keyhole Dynamics," *Phys. Rev. Applied*, vol. 11, p. 064054, 2019.
- [11] W. Huang, H. Wang, T. Rinker and W. Tan, "Investigation of metal mixing in laser keyhole welding of dissimilar materials," *Materials and Design*, p. 109056, 2020.
- [12] "FLOW-3D | Solving the world's toughest CFD problems," Flow Science Inc. , [Online]. Available: [www.flow3d.com](http://www.flow3d.com).
- [13] C. Hirt and B. Nichols, "Volume of fluid (VOF) method for the dynamics of free boundaries," *Journal of Computational Physics*, vol. 39, no. 1, pp. 201-225, 1981.
- [14] R. W. Schrage, A Theoretical Study of Interphase Mass Transfer, Columbia University Press, 1953.
- [15] M. Bayat, S. Mohanty and J. H. Hattel, "Multiphysics modelling of lack-of-fusion voids formation and evolution in IN718 made by multi-track/multi-layer L-PBF," *International Journal of Heat and Mass Transfer*, vol. 139, pp. 95-114, 2019.
- [16] V. Semak and A. Matsunawa, "The role of recoil pressure in energy balance during laser materials processing," *J. Phys. D: Appl. Phys*, vol. 30, pp. 2541-2552, 1997.
- [17] W. E.King, H. D.Barth, V. M.Castillo, G. F.Gallegos, J. W.Gibbs, D. E.Hahn, C. Kamath and A. M.Rubenchik, "Observation of keyhole-mode laser melting in laser powder-bed fusion additive manufacturing," *Journal of Materials Processing Technology*, vol. 214, pp. 2915-2925, 2014.
- [18] [Online]. Available: [https://www.engineeringtoolbox.com/convective-heat-transfer-d\\_430.html](https://www.engineeringtoolbox.com/convective-heat-transfer-d_430.html).
- [19] S. Software, "Sente Software - JMatPro," [Online]. Available: <https://www.sentesoftware.co.uk/jmatpro>. [Accessed 19 January 2023].
- [20] M. J. Torkamany, M. J. Hamed, F. Malek and J. Sabbaghzadeh, "The effect of process parameters on keyhole welding with a 400W Nd :YAG pulsed laser," *Journal of Physics D: Applied Physics*, vol. 39, no. 21, 2006.
- [21] F. M. Ghaini, M. Hamed and M. T. a. J. Sabbaghzadehb, "Weld metal microstructural characteristics in pulsed Nd: YAG laser welding," *Scripta Materialia* 56, vol. 56, pp. 955-958, 2007.
- [22] D. Bergström, J. Powell and A. Kaplan, "The absorptance of non-ferrous alloys to Nd:YAG and Nd:YLF laser light at room temperature," *Applied Optics*, vol. 48, no. 8, pp. 1290-1301, 2007.
- [23] D. Bergström, A. Kaplan and J. Powell, "The Absorptance of Steels to Nd:YLF and Nd:YAG Laser Light at Room Temperature," *Applied Surface Science*, vol. 253, no. 11, pp. 5017-5028, 2007.